AgentBreak : The Autonomous AI Red Team

Backstory: AI agents are increasingly being trusted with tools, private records, external services, and multi-step decision-making. However, these capabilities introduce new security risks. A malicious instruction hidden inside a document, a poorly configured tool permission, or a vulnerability in an agent's memory can cause it to perform actions beyond its intended purpose. Traditional security testing often relies on predefined attack scenarios, making it difficult to uncover complex vulnerabilities that emerge as an agent interacts with its environment. Problem Statement: Build AgentBreak, an autonomous AI security testing platform that actively probes AI agents to discover exploitable vulnerabilities in their instructions, memory, retrieved content, tool permissions, and multi-step execution. The platform must generate reproducible attack traces, demonstrate vulnerabilities within a controlled sandbox, and evaluate whether security defenses can prevent the same attacks without breaking legitimate agent functionality. Key Features: Autonomous Attack Discovery: Identify vulnerabilities through adaptive adversarial testing rather than relying exclusively on predefined attack cases. Multi-Stage Attack Analysis: Trace how malicious inputs, retrieved instructions, and agent decisions can combine to produce unauthorized actions. Tool and Permission Auditing: Detect situations where an agent can access data or execute actions beyond its intended permissions. Reproducible Attack Traces: Record the inputs, intermediate decisions, tool calls, and outcomes required to reproduce a discovered vulnerability. Defense Validation: Evaluate proposed mitigations against previously successful attacks and measure whether they prevent exploitation. Security Evaluation Dashboard: Visualize attack paths, vulnerability severity, exploit success rates, and defense effectiveness. Regression Testing: Check whether security improvements introduce new vulnerabilities or interfere with legitimate agent behavior.